CS 473 (Spring 2025) Yiming Qin (yimingq3@illinois.edu)

Problem 9.2

Solution:

Lemma. Let s be a square in the plane, and on three distinct sides of s choose three disjoint open segments, **none of which touches a corner of** s. Then there exists a triangle whose edges **exactly** coincide with those three given segments.

Proof. On Side i let the open segment be $[u_i, v_i]$, oriented so that as we traverse the boundary of s in clockwise order we meet u_i before v_i. Since all three segments have the same length and lie on distinct sides, the six points

$$u_1,\ v_1,\ u_2,\ v_2,\ u_3,\ v_3$$

appear in convex position along the boundary of s.

Consider the three lines

$$\ell_1 = \overline{v_1 u_2}, \quad \ell_2 = \overline{v_2 u_3}, \quad \ell_3 = \overline{v_3 u_1}.$$

We claim that no two of these lines are parallel.

Suppose, for sake of contradiction, that ℓ_i and ℓ_j were parallel. Because each segment $[u_k, v_k]$ lies on a different side of the square, the only way, say, $\overline{v_1 u_2} \parallel \overline{v_2 u_3}$ can occur is if those lines both lie in directions parallel to one pair of opposite sides of the square. But then the third line $\ell_3 = \overline{v_3 u_1}$ must connect points on the remaining pair of opposite sides, and so ℓ_3 would pass through a corner of the square. This contradicts the segment construction.

Hence ℓ_1, ℓ_2, ℓ_3 are pairwise non-parallel, and so they form the three sides of a triangle T.

Reduction Construction

Idea: We reduce from Vertex-Cover on graphs with maximum degree 3. Given a graph $G = (V, E)$ with $\Delta(G) \leq 3$ and an integer K, we construct an instance of Covering-Square-by-Triangles.

1. Since $\Delta(G) \leq 3$, the graph is 4-edge-colorable. Compute in polynomial time a proper 4-edge-coloring

   $$c(e) = j,\ j \in \{1, 2, 3, 4\},\ e \in E.$$

   Let

   $$k \ = \max_{j \in \{1,2,3,4\}} |\{c(e) = j\}|,\ e \in E.$$

2. Label the four sides of a square s by $1, 2, 3, 4$. On side j partition its interior into $k + 2$ equal open segments

   $$S_{j,1},\ S_{j,2}, \ldots, S_{j,k+2}\,.$$

   Since each color class has at most k edges, we can assign each edge $e \in E$ of color j to a distinct segment $S_{j,i}$. The remaining two segments covering vertices of the square stay unused.

3. For each vertex $v \in V$ with incident edges $e_1, \dots, e_d$ ($d \leq 3$). By the lemma, there is a unique triangle T_v whose edges exactly cover $S_{c(e_d),i}(d \leq 3, i \in \{1, 2, \cdots, k+2\})$. (If $d < 3$, the triangle is easier to construct by adding some dummy vertices near the center of the square.)

   Collect all such $\{T_v : v \in V\}$.

4. Any segment $S_{j,i}$ not used by an edge gets its own "fill-in" triangle that exactly covers that single segment. Call the set of all these triangles F.

   Let
$$T = \{ T_v : v \in V \} \cup F, \qquad W = K + |F|.$$

Correctness of the Reduction

($\Rightarrow$) If G has a vertex cover of size K:

Let $V' \subseteq V$ be a vertex cover of size K. Then select triangles
$$Q = \{T_v : v \in V'\} \cup F,$$
which has total size $K + |F| = W$.

For every edge $e = (u, v)$, at least one of u or v is in V', so the corresponding δ_e is covered by the triangle T_u or T_v. All remaining parts of the square boundary are covered by triangles in F, so Q covers the entire boundary.

($\Leftarrow$) If $Q \subseteq T$ of size k covers the square boundary:

Since the fill-in triangles in F are the only ones that cover the leftover segments, all of F must be included in Q.

Thus, $Q \setminus F$ has size at most K and must consist of triangles from T_V. Let $V' = \{v \in V : T_v \in Q\}$. By the Lemma, each edge is exactly covered by some triangles, and for every edge $e \in E$, the segment Δ_e must be covered, so some T_v of v incident to e must be in Q. Therefore, V' is a vertex cover.

Complexity

Note that the size of the constructed instance is polynomial in the size of the original graph. Each edge maps to a small interval on the square boundary, and each vertex yields at most one triangle. The number of fill-in triangles is smaller than $|E|$, since they only cover the leftover segments between δ_e. Hence, the total number of triangles in T is $|V| + O(|E|)$, and all geometric constructions (edge coloring, triangle placement) can be done in polynomial time. Therefore, the reduction is polynomial-time and COVERING-SQUARE-BY-TRIANGLES is NP-hard.